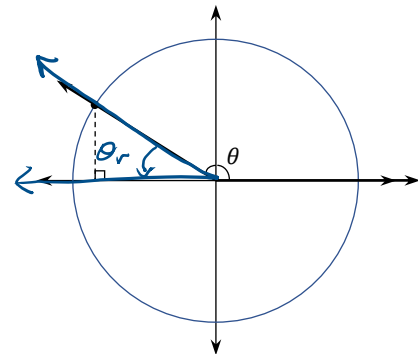


Evaluating Trigonometric Functions with Reference Angles

Reference Angles

For an angle θ , in standard position, we often consider a triangle with hypotenuse along the terminal side of θ and base along the x -axis. A right triangle formed in this way is frequently referred to as a reference triangle.

The acute angle formed by the hypotenuse and the x -axis is called the reference angle. This angle is typically denoted as θ_r or θ' .

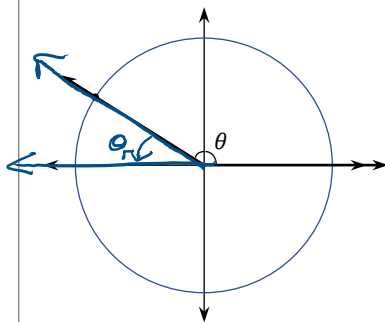


DEFINITION: Reference (Related) Angles

For a non-quadrantal angle θ , in standard position, the _____ angle is the positive acute angle formed by the terminal side of θ and the x -axis.

EXAMPLE 1

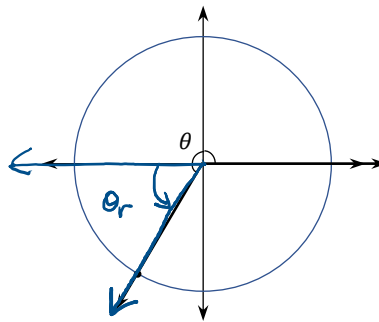
For each angle θ below, provide a reasonable estimate for θ , and determine the associated reference angle.



Answers vary...

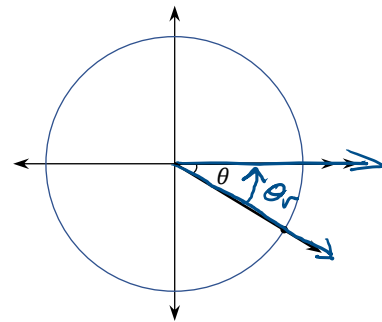
$$\theta \approx 140^\circ$$

$$\theta_r \approx 40^\circ$$



$$\theta \approx 240^\circ$$

$$\theta_r \approx 60^\circ$$



$$\theta \approx -40^\circ$$

$$\theta_r \approx 40^\circ$$

PROCEDURE: Finding the Reference Angle for any Non-Quadrantal Angle

Given a non-quadrantal angle θ , in standard form, θ_r can be found using the following procedure.

Step 1: If $\theta > 360^\circ$ or if $\theta < 0^\circ$, find a coterminal angle by adding or subtracting 360° as many times as needed, to get an angle greater than 0° , but less than 360° .

Step 2: If the result from Step 1 is between 0° and 90° , then this result is your reference angle. If not, subtract 180° and continue to Step 3.

Step 3: If the result from Step 2 is between 0° and 90° , then this result is your reference angle. If this result is between -90° and 0° , then the absolute value of this result is your reference angle. If not, subtract 180° and repeat this step.

Note: You should take some time to understand *WHY* this procedure works.

EXAMPLE 2

Given θ , find θ_r .

(a) $\theta = -32^\circ$

$$\theta_r = 32^\circ$$

(b) $\theta = 197^\circ$

$$\begin{array}{r} -180 \\ \hline 17 \end{array} \rightarrow \theta_r = 17^\circ$$

(c) $\theta = 288^\circ$

$$\begin{array}{r} -180 \\ \hline 108 \\ -180 \\ \hline -72 \end{array} \rightarrow \theta_r = 72^\circ$$

(d) $\theta = 682^\circ$

$$\begin{array}{r} -360 \\ \hline 322 \\ -180 \\ \hline 142 \\ -180 \\ \hline -38 \end{array} \rightarrow \theta_r = 38^\circ$$

(e) $\theta = -730^\circ$

$$\begin{array}{r} +360 \\ \hline -370 \\ +360 \\ \hline -10 \end{array} \rightarrow \theta_r = 10^\circ$$

(f) $\theta = 900^\circ$

$$\begin{array}{r} -360 \\ \hline 540 \\ -360 \\ \hline 180 \end{array} \rightarrow \theta = 900^\circ \text{ is a } \underline{\text{quadrantal angle}}$$

$\theta_r \text{ DNE}$

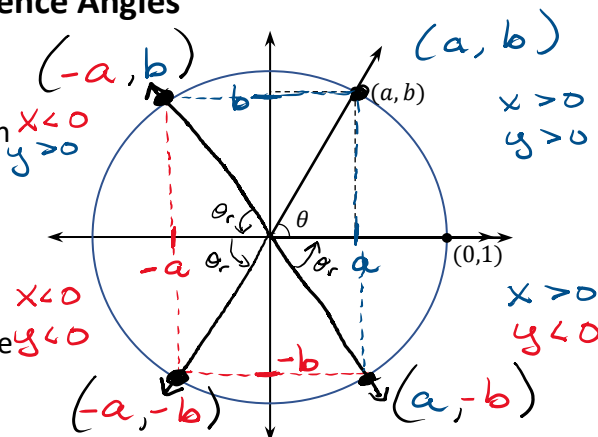
Evaluating Trigonometric Functions with Reference Angles

Consider the angle θ shown here.

Construct additional angles, one for each quadrant, with the same reference angle as θ .

What do you notice about the point where the terminal side for each of these angles intersects the circle?

Use this visual to better understand the Reference Angle theorem below.



THEOREM: Reference Angle Theorem

Let θ be any non-quadrantal angle, in standard position, with reference angle θ_r , and let f be any trigonometric function, then $|f(\theta)| = f(\theta_r)$.

That is, $f(\theta)$ is equal to $f(\theta_r)$, except for a possible change in sign. This sign change is determined by the quadrant for the angle θ .

Note: This issue in sign can be resolved using the mnemonic, "All Students Take Calculus."

EXAMPLE 3

** post QI table on board for reference*

Evaluate each expression using reference angles.

(a) $\cos(210^\circ)$

$\theta = 210^\circ \in \text{QIII}$
 $\cos(210^\circ) < 0$

$\theta_r = 30^\circ$

$\cos(210^\circ) = -\cos(30^\circ) = -\frac{\sqrt{3}}{2}$

(b) $\sin(480^\circ)$

$\theta = 480^\circ \in \text{QII}$
 $\begin{array}{r} -360 \\ \hline 120^\circ \end{array}$ coterminal

$\sin(480^\circ) > 0$

$\theta_r = 60^\circ$, $\sin(480^\circ) = \sin(60^\circ) = \frac{\sqrt{3}}{2}$

(c) $\tan(765^\circ)$

$\theta = 765^\circ \in \text{QI}$
 $\begin{array}{r} -360 \\ \hline 405 \\ -360 \\ \hline 45^\circ \end{array}$ coterminal

$\tan(765^\circ) > 0$, $\theta_r = 45^\circ$

$\tan(765^\circ) = \tan(45^\circ) = 1$

(d) $\csc(-225^\circ)$

$\theta = -225^\circ \in \text{QII}$
 $\begin{array}{r} +360 \\ \hline 135^\circ \end{array}$ coterminal

$\csc(-225^\circ) > 0$

$\theta_r = 45^\circ$

$\csc(-225^\circ) = \csc(45^\circ) = \frac{1}{\sin(45^\circ)}$
 $= \frac{2}{\frac{\sqrt{2}}{2}} = \sqrt{2}$

S	A
T	C

EXAMPLE 4** post QI table on board for reference*

Let θ be an angle in standard position, with $0^\circ \leq \theta < 360^\circ$. Find all possible θ -values that satisfy each trigonometric expression.

(a) $\cos(\theta) = -\frac{\sqrt{3}}{2} < 0$ thus $\theta \in \text{QII or QIII}$

S	A
T	C

$$\cos(30^\circ) = \frac{\sqrt{3}}{2} \text{ thus } \theta_r = 30^\circ$$

$$\theta = 180^\circ - 30^\circ = 150^\circ$$

or

$$\theta = 180^\circ + 30^\circ = 210^\circ$$

(b) $\tan(\theta) = \sqrt{3} > 0$ thus $\theta \in \text{QI or QIII}$

$$\tan(60^\circ) = \sqrt{3} \text{ thus } \theta_r = 60^\circ$$

$$\theta = 60^\circ$$

or

$$\theta = 180^\circ + 60^\circ = 240^\circ$$

(c) $\csc(\theta) = \sqrt{2} \rightarrow \sin \theta = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} > 0$ thus $\theta \in \text{QI or QII}$

$$\sin(45^\circ) = \frac{\sqrt{2}}{2} \text{ thus } \theta_r = 45^\circ$$

$$\theta = 45^\circ$$

or

$$\theta = 180^\circ - 45^\circ = 135^\circ$$

Looking Ahead...

Reference angles can be helpful tools for evaluating trigonometric functions. However, we'll soon discover an even better resource for finding exact trigonometric values, the **unit circle**. The construction of the unit circle will take some practice, but this effort will produce a more efficient and reliable way to evaluate these expressions.

